Matching Evidence to Decisions in Prolonged Crises

Can we say beforehand how far a finding travels, and for which decision?

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> **Example of the decision we study (hypothetical).** After repeated drought, should a programme extend short-term cash transfers or link households to a long-term safety net?

Why these decisions matter: crises are large and last for years

- Fragile and conflict-affected settings hold a growing share of the world's extreme poor (World Bank, 2025 projection)
- About seven in ten refugees are in protracted situations (UNHCR, 2025)
- About 239 million people need humanitarian aid (OCHA, 2026)

Who decides, what they decide, and why discretion matters

Topic of study: the **humanitarian–development nexus**, the established term for coordinating life-saving aid with longer-term development when shocks repeat or crises last for years.

VISUAL: the decision landscape. Two columns of decisions, with the actors who make them: *immediate response* (who gets emergency aid next month; mostly decided in the field, and where most existing decision studies sit) versus *nexus design* (how to sequence and link aid, safety nets and livelihood support over years; shaped by headquarters, donors and evidence funders as well as field planners; **this agenda**). Rules, budget and mandate close in from the sides, leaving a window labelled "planner discretion".

- **Planner discretion** is the choice left once rules, budget, mandate and norms have fixed what they fix
- Evidence can only change choices that are open: in the example, extending cash or linking to a safety net is only a real choice if the rules and budget allow both
- So we **measure discretion** decision by decision instead of assuming it

Evidence is worth seeking only if it could change the choice

VISUAL: a decision flow. A decision enters; three tests in sequence: *room to choose?* (no: set aside, outside scope) then *uncertainty about the best option?* (no: no evidence needed) then *relevant evidence exists?* Outcomes branch: **use it** (relevant and usable), **gap: transport it** (exists but too distant), **gap: generate or substitute** (none). The research target is the share of decisions ending in each outcome, by sector and decision type.

- **Value of information** comes from health economics: more evidence is worth getting only if it could change which option is chosen. We apply it to programme design
- Relevance has two parts: **distance** (will it hold here?) and **actionability** (can this planner implement it?)
- A real case: the Kenya HSNP Phase 2 business case weighed seven expansion options. The evidence supported all of them and did not separate them

Where this sits in the literature

VISUAL: five fields around one decision, each marked with what it gives and the gap it leaves. Generalizing from impact evaluations (effects vary by setting; mostly stable settings); transportability and data fusion (when an effect can be carried over; rarely tested on real programme decisions); evidence use and uptake (why evidence is not used; seldom measures discretion); evidence synthesis and gap maps (where studies exist; not whether relevant); decisions under uncertainty (value of information, adaptive pathways; not applied to auditing causal evidence). The decision at the centre is the unfilled gap.

Inset: a 3ie evidence gap map. A matrix of interventions (rows) by outcomes (columns) for one topic. Each cell shows how many studies exist, so empty cells are the unknowns.

- Nearest humanitarian work found: IDinsight (value of information for commissioning an evaluation), ALNAP (immediate-response decisions), Elrha (barriers). None audits existing evidence against a nexus design decision
- **Why start in crises:** the hard case has the least evidence and the most constraint, so a method that works here should carry to fields with more and closer evidence

A puzzle and an aspiration: two Kenyan cash studies disagree, and one programme travelled to conflict

VISUAL: two panels.

Left, same country, two answers. *Egger et al.*: a randomized trial of **one-time cash transfers** to poor rural households in Kenya, measuring spillovers to the local economy; reports a local multiplier of about 2.5 (income created per unit transferred). *HSNP Phase 2 evaluation*: **regular, unconditional cash every two months** to chronically poor households in four arid northern counties, with a local-economy model of income created. Between the two, a question mark: the quantities are defined differently, and why they differ is not established. Both are on our replication list.

Right, evidence that travelled. *Bedoya et al., Afghanistan*: a **graduation programme**, a bundle of a livestock asset, a cash stipend, skills training and coaching for the ultra-poor, first shown to work in stable settings and then tested in conflict; pre-registered; consumption up about 30 percent. An arrow from a "stable settings" tile to a "conflict setting" tile. Caption: what made it possible? (No replication data found.)

Evidence–decision mismatch for humanitarian–development planners: problem, questions, goal

> Evidence and research attention are largely **non-market goods**, so what gets studied can be misallocated, while planners still lack evidence they can use. The mismatch is one of relevance.

Discretion and value of evidence

How much discretion do nexus planners have over a decision, and how useful do they judge the evidence available to them?

Decision mapping

What informs planners' decisions today (context data, tacit knowledge, best practice, standard procedures), and what keeps relevant evidence from reaching them: difficulty finding it, too much of it, or evidence that does not fit their setting?

Evidence transport

Which barriers to using existing evidence (search cost, scattered studies, studies from settings too different) can be reduced? How well can we predict the effect of an intervention in a new setting, setting realistic targets that are checked against outcomes afterwards?

Primary goal: better-designed interventions. Checking whether predictions were accurate serves that goal; it is not the goal itself.

Sub-goals: more efficient allocation of research funding; more targeted evidence generation by development organizations; an open-source toolkit that generalizes to other uses.

Discretion and value of evidence: what we measure first

- Four questions per decision: is there **discretion left** after constraints? Can **existing information** already separate the options? Does the **timeline** allow more? Who bears the **cost of being wrong**, and is it reversible?
- **First instrument: a desk audit.** A review of the full documentation of one evaluated intervention, analysing the decisions made, the constraints, the evidence used and the outcomes. No human subjects
  - includes whether decision-makers sought **outside support** for evidence, and under what conditions (revealed demand)
- **Later:** a documented partner funnel (approached, responded, accepted, fit), where refusals are data
- In scope: headquarters versus field constraints; where evidence generation should be allocated

> **Still open:** the taxonomy of decisions (immediate response, long-term planning, operational, design), and the headquarters/field relationship.

Decision mapping: what informs decisions today

- Plans are informed by **context data, tacit knowledge, best practice and standard procedures**
- The unit of study is the **assumptions** behind a decision, in causal maps, scenarios and log-frames (the logic tables programmes write), stated or not
- Three barriers to evidence reaching the planner:
  - **search:** relevant evidence is hard to find
  - **overload:** a lot of evidence, few strong signals
  - **relevance:** the evidence available does not fit the planner's setting

> **Feeds evidence transport.** The causal chain, blockers and log-frame assumptions become the input to the transport model: they say which links a source study must be checked on, and the planner reviews them at each staged deliverable.

Eliciting causal maps from practitioners is the established field of **systems mapping**, usually qualitative. Linking it to causal graphs for quantitative transport is the untested part.

Evidence transport: how far does a finding travel? What counts as comparable

- We predict the effect of an intervention in a new setting from studies elsewhere, to set a realistic target, then check the prediction against the measured outcome afterwards
- Source and target differ along **named dimensions:** spatial (including environment), temporal, institutional, market, treatment version, population, interference
- **Comparable is a profile, not one score.** On each dimension that the planner's causal map says matters:
  - **Objective:** a measured distance, for example the two areas' vegetation index has a similar distribution
  - **Subjective:** the planner's similarity rating, for example forest health 4 on a 6-point scale
- Comparable for a decision when both fall inside thresholds the planner accepts; a gap between the two is flagged
- Idea for choosing cases: an insecure region may resemble an insecure region elsewhere more than the rest of its own country

The scale is illustrative. More than one expert rating is ideal, subject to availability.

The pipeline already under construction

VISUAL: a 2×2 with a loop, coloured by maturity. Columns: *find and describe what is known* and *test what can be used here*. Rows: *evidence from studies* and *local, replicable data and results*.

- **Literature discovery and extraction** (find; studies): automated search, structured extraction of study facts and causal claims. *Built and piloted*
- **Evidence gap map** (test; studies): the **3ie gap map** model, extended with context (shock, crisis phase, delivery institution, displacement, horizon). *In design*
- **Context data panel** (find; local): conflict, rainfall, prices, displacement and boundaries fused through open-source APIs and scripted or agent-assisted fetching; new country by configuration. *Built: Somalia and Sudan*
- **Reproducible replications,** `reprokit` (test; local): extract and replicate eligible studies. *Released; Bangladesh insurance replication nearly complete in full; next: Egger and HSNP Phase 2*

Arrows: the gap map directs the search; extraction makes comparable records; the context panel measures closeness; replication checks which estimates reproduce; results update the map.

- `reprokit` records hashes and rebuilds in continuous integration, so anyone can re-run a result
- Replication is fast when geodata are public, or when we replicate without them; slow only if locations must be requested privately
- A **proof of concept for a re-runnable, open-source pipeline**, validated in the dissertation on one applied case

The transport model: built with the decision-maker

1. **Decision mapping supplies the target map:** causal chain, blockers, log-frame thresholds
2. **Match the maps:** the source study's causal story is matched to the planner's, so both use the same variables
3. **Flag the links that differ** between the source and target causal graphs
4. **Measure and judge:** context data give the objective distance; the planner gives the similarity rating
5. **Compute a prediction** of one of four types
6. **Translate** to the planner's area, population and scale, as a target metric they can check; they correct the map and the cycle repeats

Four prediction types

Type	When
Point estimate with standard errors	Maps match, distance small
Bounds (Gechter)	Partial overlap
Ordinal category or sign	Only direction supported
"Insufficient evidence" or "distance too great"	No defensible transport: a result

> **Not a black-box predictor.** The person who will act on the answer judges what is close enough, using a map they helped draw.

Which differences matter: selection diagrams

A **selection diagram** is a causal graph, with boxes for variables and arrows for cause and effect, plus a mark on each arrow that works differently in the source and the target setting.

VISUAL: one small causal graph drawn twice, source and target. For cash transfers: *transfer* → *household spending* → *local sales* → *local income*, with side inputs *market access* and *security*. Marked arrows in the target (for example spending → local sales, because markets are thinner; security → local sales). Unmarked arrows are labelled "carries over"; marked ones "measure before transporting".

- Marked arrows say which differences **must be measured**, and whether a way to carry the result over exists at all
- Here the graph comes from the planner's decision map, matched to the source study's causal story

Approaches to explore, and the context data behind them

Group	Approach, in plain words	Where the expert enters
What can be carried over?	**Selection diagrams:** which causal links differ	The graph comes from the planner's map
How to combine evidence	**Hierarchical Bayesian models:** borrow strength across sites by estimating how much effects vary between them	Priors; which sites count as comparable
How to combine evidence	**Similarity-weighted borrowing:** weight studies by how close their setting is to the target	Which variables, and how close is close enough
What if it cannot?	**Partial-identification bounds:** report the range of effects consistent with what is known	The assumptions the bounds rest on

- **Context data:** remote sensing (rainfall, vegetation indices from Landsat, MODIS and Sentinel-2, night lights) plus conflict events, prices, displacement, boundaries
- **Limits:** satellites describe environment and activity, not treatment or welfare; resolution must suit the unit; site coordinates decide what can be linked
- **Test:** the Egger versus HSNP pair. Do the approaches agree on whether a result carried from one setting to the other?

Progress so far and what comes next

Done so far

- Initial novelty check of the three questions against the nearest prior art (a spot check)
- Literature discovery and extraction built and piloted; context data panel built for Somalia and Sudan; `reprokit` released
- Replication of the Bangladesh insurance study nearly complete in full
- Next replication candidates identified (Egger, HSNP Phase 2); papers transcribed for direct reading

Next

- **Desk audit** of one evaluated intervention: its full documentation, analysed for the decisions made, the constraints, the evidence used and the outcomes. No human subjects
- Replicate and compare the two Kenyan cash studies
- Extend to the evidence gap map and to more replications
- **Outreach to J-PAL and 3ie** on decisions and evidence gaps in crisis settings, and on replication data access

Timeline: three gates on the thesis schedule

VISUAL: a horizontal timeline, Nov 2026 to spring 2029, with the three gates as markers.

- **Nov 2026:** peer talk (now)
- **Apr 2027:** faculty talk
- **May 2027, gate 1:** faculty approval and pre-registration of the discretion audit and the plan
- **Jun 2027:** IRB submission
- **Fall 2027 to spring 2028:** recruit potential nexus expert decision-makers (human subjects)
- **Summer 2028, gate 2:** field work; real causal mapping with practitioners
- **Fall 2028 (year 3):** build and validate the causal transport model through iterative work with a partner
- **Spring 2029, gate 3:** thesis paper complete

Running along the top, bars: desk audit and replications (now to spring 2027); outreach to J-PAL and 3ie (now onward).

What the gates are for. Each is a point where the work must show it can deliver before the next, costlier step, so we stop or redirect early rather than late:

- **Gate 1, is the question answerable?** An approved audit shows decisions, constraints and evidence use can be measured from documents
- **Gate 2, can decisions be mapped?** Practitioners' causal maps are usable and the planner finds them useful
- **Gate 3, does transport work?** A method with empirical validation, then a partner uses it; predictions are checked against outcomes

Dates are the owner's current plan (2026-11-13), not commitments. Practitioner work follows IRB approval.

Appendix

A1. What a selection diagram can and cannot do

- It tells you which differences you **must measure** and whether a **formula to carry the result over exists at all**
- It **cannot** create data that do not exist, repair a missing outcome, or make an impossible delivery possible
- The formalism is Bareinboim and Pearl's; the graph here comes from the planner's decision map, matched to the source study's causal story

A2. The full decision table

Relevant evidence exists?	Discretion: yes	Discretion: no (fixed by rules, budget, mandate)
Yes: relevant and actionable	**Use it**	On the shelf
Yes, but too distant	**Gap: transport the evidence (RQ3)**	Not a priority
Yes, but not actionable here	**Gap: redesign or adapt (RQ2)**	Not a priority
None	**Gap: generate or substitute**	Ignore

- Scope: decisions where the planner already has discretion and lacks relevant evidence; evidence loosening constraints is plausible but out of scope
- Other states: little demand beyond logistics; abundant evidence not used; evidence used well; planners make do; insufficient evidence either way

A3. Comparability profile (illustration)

Dimension	Objective measure	Subjective rating (1 to 6, illustrative)
Environment	Overlap of the vegetation-index distributions	Planner: forest health 4
Market	Market-access or price-pass-through indicator	Planner: 3
Institution	Delivery channel (NGO, public safety net, payment rail)	Planner: 5
Conflict exposure	Event counts and trends	Planner: 2
Population	Displacement status, composition	Planner: 4

- Thresholds per dimension are set with the planner; a gap between the two columns on a row is flagged for discussion
- Values here are placeholders, not data

A4. Validation, gates and timeline

Question	Validated when
RQ1	A working paper on the knowledge audit is approved by faculty, and at least one fit partner agrees with the funnel documented
RQ2	Desk phase: documents suffice to model constraints, scenarios and log-frames. Partner phase: the partner finds the process and products useful (needs ethics approval)
RQ3	Partial: a methodology paper with empirical validation. Full: a partner uses the results. Follow-on: predictions turn out reasonably correct

- **Dates (owner's plan, 2026-11-13):** faculty talk April 2027; faculty approval and pre-registration May 2027; IRB submission June 2027; recruitment fall 2027 to spring 2028; field work summer 2028; transport model validated with a partner fall 2028; thesis paper spring 2029
- **Case-level checks before investing in a case:** case and partner fit; is new evidence worth it; what can be inferred; can it be evaluated

A5. Bias controls for the feedback rounds

Someone will ask how planner feedback, with variables not fixed in advance, avoids p-hacking. The top ideas:

1. **A written, registered protocol:** the steps (data discovery, map matching, comparability profile, modelling, planner review, repeat), a stopping rule, fixed dimensions and approaches, and a log of every round and variable tried
2. **The planner rates similarity before seeing the predicted effect;** changes made after seeing the estimate are flagged
3. **Report sensitivity:** the prediction under every reasonable specification tried
4. **Lock the prediction before the outcome** (pre-registered, time-stamped)

Further controls (a planner-free baseline; more than one expert where available) are for the protocol to settle.

A6. Project maturity: built, planned, gated

Component	Status
Context panel (Somalia, Sudan)	Built and reproducing; separate config per country
`reprokit`	Released (v0.4.0, evidence kind added)
Literature discovery and extraction	Built and piloted
Evidence map (3ie-style gap map)	In design; study catalogue first
Bangladesh insurance replication	Nearly complete in full (owner, 2026-11-13); village coordinates are the gate for the satellite-linked step
Transferability	Planned; no repository yet
Data-fusion toolkit	Deferred

A7. Claims we keep, soften or drop

Claim	Treatment
"No one has audited evidence for programme design"	**Soften:** nearest prior art found
Egger versus HSNP as a transport failure	**Soften:** hypothesis-generating pair
A complete corpus of trials; universal impact ranges; a live simulator or dashboard	**Dropped**
Impact estimates in novel localities	**Soften** to a tiered claim
Participatory causal maps produce usable log-frames	**Proposal**
The compound-shocks panel supports transfer	**Narrowly:** context only
Evidence-use patterns across the field	**Qualitative only;** the scans were not systematic

A8. Positioning and dropped use cases

- **Prior art found:** Brown et al. 2025 and IDinsight (value of information for funding); ALNAP (immediate-response decisions); Elrha (barriers); Egger-style and Bedoya-style trials (transport and fragile-setting cases)
- **Dropped use cases:** anticipatory action and index insurance are examples of modality and timing questions only, not research strands of this agenda; operational logistics optimization is a separate adjacent agenda
- **Not a claim** about which intervention is best

A9. Open items to validate in the next few weeks

Item	Why	Next step
Scale figures (World Bank, UNHCR, OCHA)	Opening slide; found via a web-search summary	Gemini check against the primary reports; name the poverty line
Egger versus HSNP definitions	Egger's multiplier is GDP per unit transferred; HSNP's 0.38 to 0.93 may be *additional* income beyond the transfer (total about 1.38 to 1.93)	Read the HSNP local-economy methods; harmonize before any comparison
Replication feasibility	Egger's package is public but village locations are restricted; HSNP data held locally; Bedoya data not found	Confirm what each supports; ask authors only for what is needed
Bedoya et al.	Aspirational case; claims checked on the abstract and registration only	Full read; check what enabled success
Stage-2 extraction	The pipeline slide says it extracts causal claims; pilot figures not re-verified	Confirm against its status doc
Literature map (slide 4)	Row descriptions are my characterization	Read one paper per field
Evidence-use scans	The "mixed, sector-dependent" picture rests on about 31 sources I have not read	Read a sample before relying on it
Nexus definition	Mine, not sourced	Read the OECD DAC recommendation
Ethics and outreach	J-PAL contact could count as human-subjects research	Ask the IRB before anything beyond public-data requests

Design questions (decision taxonomy, headquarters/field relationship, the definition of "comparable") are tracked separately.

A10. Sources consulted (1 of 2): read in full or checked against the text

- **Programme documents:** FCDO, Kenya Hunger Safety Net Programme Phase 2 — Business Case and Completion Review
- **Egger, Haushofer, Miguel, Niehaus, Walker (2022).** General equilibrium effects of cash transfers: experimental evidence from Kenya. *Econometrica*
- **Bedoya, Coville, Haushofer, Isaqzadeh, Shapiro (2019).** No household left behind: Afghanistan Targeting the Ultra Poor impact evaluation. NBER WP 25981
- **Pritchett and Sandefur (2015).** Context matters for the size of impact: experiments in other settings (microcredit)
- **Hjort et al. (2021).** How research affects policy: Brazilian municipalities
- **Crost, Felter and Johnston (2014; 2016).** Aid, insurgent response and conflict, Philippines (community-driven development; conditional cash transfers)
- **Brown, Salisbury and Ball (2025).** Value of information applied to grant allocation
- **Groh et al. (2016); Tennant et al. (2021); Altmejd et al. (2019).** Checked against the transcripts for specific claims

Full-text transcripts are held locally (git-ignored); identifiers and DOIs are in the novelty-check notes.

A11. Sources consulted (2 of 2): identifier-matched, secondhand or not yet read

- **Matched to a registry, not read:** Vivalt (2020); Allcott (site selection bias); Meager (Bayesian hierarchical analysis of microcredit); Banerjee et al. (2016, proof of concept to scalable policies); Oliver et al. (2014, barriers to evidence use); Snilstveit et al. (2016, evidence gap maps); Kwakkel, Haasnoot and Walker (2016, robust decision-making and adaptive pathways); Bareinboim and Pearl (2012, transportability)
- **Read by Gemini only:** Gechter (bounds); Montiel Olea et al. (site selection); Bongers et al. (cyclic causal models); DellaVigna, Pope and Vivalt (2019); Kastner et al. (2009) and White et al. (2020) on capture-recapture; Bold et al. (2018); IDinsight (2022); ALNAP, Campbell and Knox Clarke (2019); Elrha, Smith, Swithern and Harrison (2022)
- **Two non-systematic scans of evidence use** (about 31 sources) by Gemini: not read by me; no figures used
- **Scale figures** (World Bank 2025; UNHCR Global Trends 2025; OCHA GHO 2026): web summaries, **being verified**
- **Not read:** OECD DAC nexus recommendation; Claxton (1999); Fenwick et al. (2008)

Reference hygiene: identifiers come from registries (Crossref), not model recall.

A12. Why it matters beyond this case

- **Development economics:** context-dependent external validity of causal treatment effects for interventions
- **Decision science:** value of information applied to programme-design decisions, with measured discretion
- **Causal inference:** transport methods tested prospectively, with pre-registered predictions, in a hard setting
- **Climate and environmental decision-making:** a decision-relevance test for adaptation methods under uncertainty
- **Research funding and synthesis:** separating real research gaps from gaps existing evidence can fill
- **Research infrastructure:** an open-source, reproducible pipeline others can re-run on a different decision
- **Humanitarian practice:** a map of what plans rest on, and where central rules block use of evidence

Intended contributions, not results. Each depends on the corresponding question working.